

Capture the Flag 4: The Student List

Goal: Build a scrollable student directory using Dart Lists and `ListView.builder`, replacing manually duplicated widgets with dynamically generated UI.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may NOT ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

Objective: Create a feature branch before building your student list.

Create:

`feature/student-list`

Screenshot:

```
C:\Users\Manes Christopher\Documents\mannes>git checkout -b feature/student-list
Switched to a new branch 'feature/student-list'
```

🚩 Flag 1 — Build Your Student Data

Objective: Store multiple students inside a Dart `List`.

```
List<Student> students = [  
  Student(  
    name: 'Armanes Christopher G. Artajo',  
    course: 'BSIT 3',  
    age: 20,  
    hobby: 'Playing Online Games',  
    image: 'assets/profile.jpg',  
    birthdate: 'Nov 16, 2005',  
  ),  
  Student(  
    name: 'Ivan Villareal',  
    course: 'BSIT 3',  
    age: 19,  
    hobby: 'VideoGrapher',  
    image: 'assets/images.jpg',  
    birthdate: 'Jan 31, 2002',  
  ),  
  Student(  
    name: 'Wilken Nobetnov',  
    course: 'BSIT 3',  
    age: 21,  
    hobby: 'Gooning',  
    image: 'assets/wklen.jpg',  
    birthdate: 'Mar 22, 2005',  
  ),  
];
```

```
),  
  Student(  
    name: 'Tonyo Moreno',  
    course: 'BSIT 3',  
    age: 18,  
    hobby: 'Roblox',  
    image: 'assets/toning.jpg',  
    birthdate: 'Aug 4, 2005',  
  ),  
  Student(  
    name: 'Joshua Santos',  
    course: 'BSIT 3',  
    age: 20,  
    hobby: 'CODE',  
    image: 'assets/shuuaa.jpg',  
    birthdate: 'Sep 3, 2005',  
  ),  
];
```

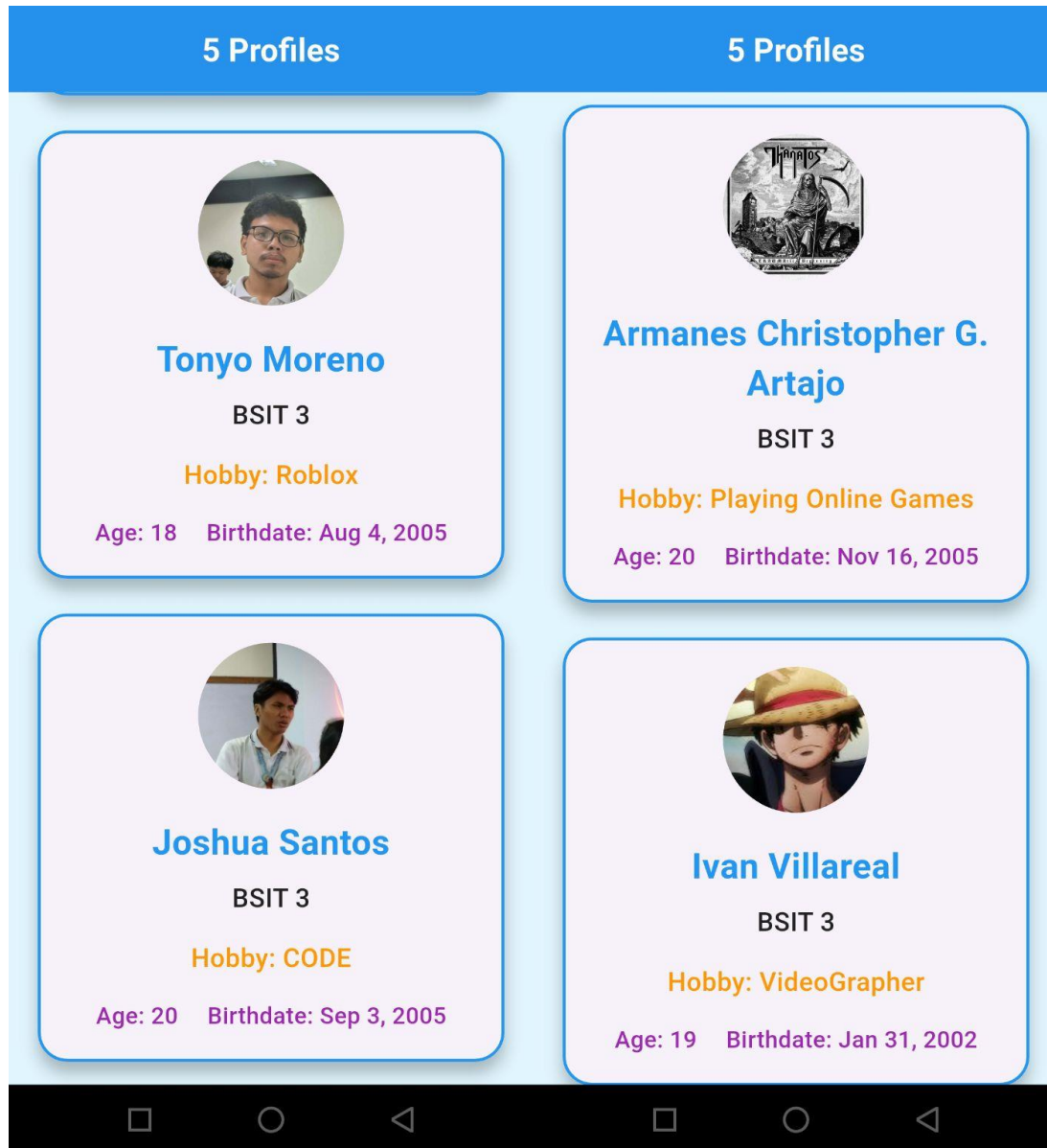
🚩 Flag 2 — The Copy-Paste Trap

 Armanes Christopher G. Artajo BSIT 3 Hobby: Playing Online Games Age: 20 Birthdate: Nov 16, 2005	 Ivan Villareal BSIT 3 Hobby: VideoGrapher Age: 19 Birthdate: Jan 31, 2002	 Wilken Nobetnov BSIT 3 Hobby: Gooning Age: 21 Birthdate: Mar 22, 2005
 Tonyo Moreno BSIT 3 Hobby: Roblox Age: 18 Birthdate: Aug 4, 2005	 Joshua Santos BSIT 3 Hobby: CODE Age: 20 Birthdate: Sep 3, 2005	

🚩 Flag 3 — Discover **ListView**

Objective: Make your student directory scrollable using **ListView**.

```
body: ListView(  
  padding: const EdgeInsets.all(16),
```

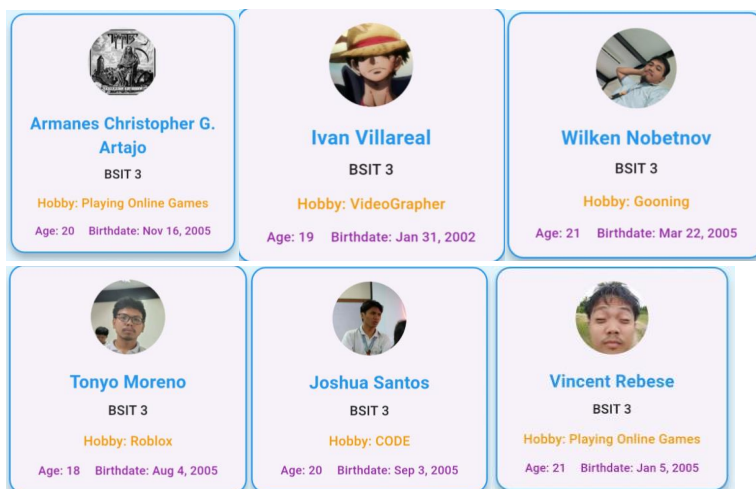


🚩 Flag 4 — Unlock `ListView.builder`

Objective: Generate student cards automatically using `ListView.builder`.

```
List<Student> students = [
  Student(
    name: 'Armanes Christopher G. Artajo',
    course: 'BSIT 3',
    age: 20,
    hobby: 'Playing Online Games',
    image: 'assets/profile.jpg',
    birthdate: 'Nov 16, 2005',
  ),
  Student(
    name: 'Ivan Villareal',
    course: 'BSIT 3',
    age: 19,
    hobby: 'VideoGrapher',
    image: 'assets/images.jpg',
    birthdate: 'Jan 31, 2002',
  ),
  Student(
    name: 'Wilken Nobetnov',
    course: 'BSIT 3',
    age: 21,
    hobby: 'Gooning',
    image: 'assets/wklen.jpg',
    birthdate: 'Mar 22, 2005',
  ),
  Student(
    name: 'Tonyo Moreno',
    course: 'BSIT 3',
    age: 18,
    hobby: 'Roblox',
    image: 'assets/toning.jpg',
    birthdate: 'Aug 4, 2005',
  ),
  Student(
    name: 'Joshua Santos',
    course: 'BSIT 3',
    age: 20,
    hobby: 'CODE',
    image: 'assets/shuaa.jpg',
    birthdate: 'Sep 3, 2005',
  ),
  Student(
    name: 'Vincent Rebese',
    course: 'BSIT 3',
    age: 21,
    hobby: 'Playing Online Games',
    image: 'assets/Binsoy.jpg',
    birthdate: 'Jan 5, 2005',
  ),
],
```

```
body: ListView.builder(
  padding: const EdgeInsets.all(16),
  itemCount: students.length,
  itemBuilder: (context, index) {
    return Column(
      children: [
        buildProfileCard(
          name: students[index].name,
          course: students[index].course,
          age: students[index].age,
          hobby: students[index].hobby,
          image: students[index].image,
          birthdate: students[index].birthdate,
        ),
        if (index < students.length - 1) const SizedBox(height: 16),
      ],
    );
  },
)
```



🚩 Flag 5 — Every Student Deserves an ID

Objective: Expand your student records with additional information.

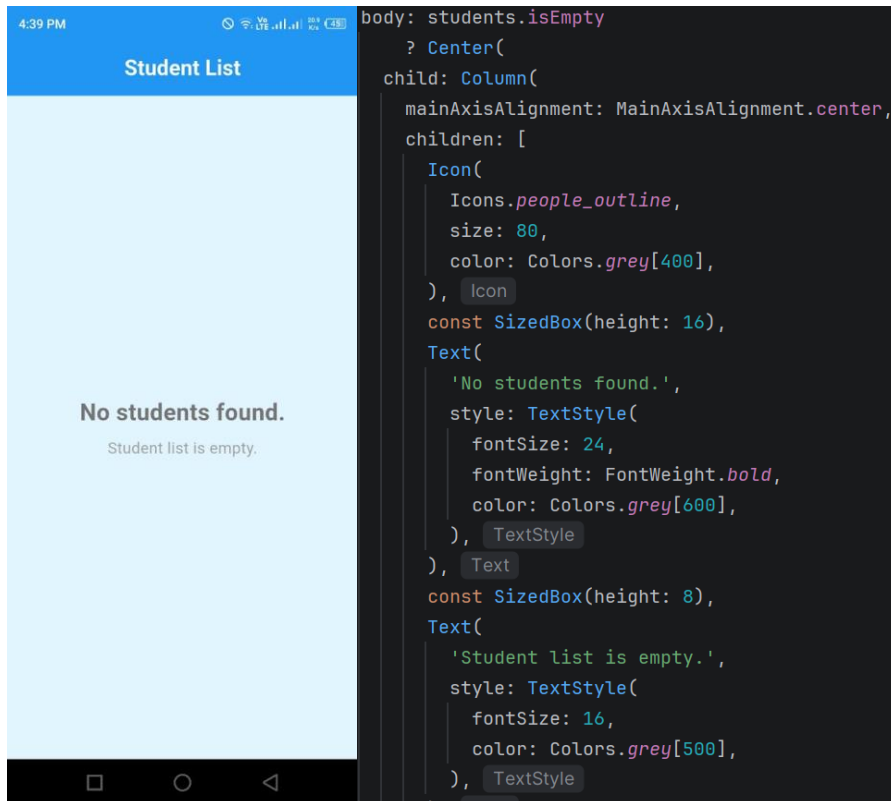
```
String studentId;  
String email;  
String favoriteSubject;
```



🚩 Flag 6 — Empty Classroom

What happens when there are no students?

Objective: Handle an empty `List` gracefully.



Flag 7 — Sort the Roll Call

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Student Data with ID	Student Data with ID	Student Data with ID	Student Data with ID
 Armanes Christopher G. Artajo BSIT 3 ID: 2024-001 Email: armanes.artajo@edu.ph Favorite Subject: IoT Hobby: Playing Online Games Age: 20 Birthdate: Nov 16, 2005	 Ivan Villareal BSIT 3 ID: 2024-002 Email: ivan.villareal@edu.ph Favorite Subject: IoT Hobby: VideoGrapher Age: 19 Birthdate: Jan 31, 2002	 Tonyo Moreno BSIT 3 ID: 2024-004 Email: tonyo.moreno@edu.ph Favorite Subject: Project Management Hobby: Roblox Age: 18 Birthdate: Aug 4, 2005	Joshua Santos BSIT 3 ID: 2024-005 Email: joshua.santos@edu.ph Favorite Subject: Project Management Hobby: CODE Age: 20 Birthdate: Sep 3, 2005
 Ivan Villareal BSIT 3	 Wilken Nobetnov BSIT 3 ID: 2024-003 Email: wilken.nobetnov@edu.ph Favorite Subject: IoT	 Joshua Santos BSIT 3 ID: 2024-005 Email: joshua.santos@edu.ph	 Vincent Rebese BSIT 3 ID: 2024-006 Email: vincent.rebese@edu.ph Favorite Subject: Project Management Hobby: Playing Online Games Age: 21 Birthdate: Jan 5, 2005

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Student Data with ID	Student Data with ID	Student Data with ID	Student Data with ID
 Armanes Christopher G. Artajo BSIT 3 ID: 2024-001 Email: armanes.artajo@edu.ph Favorite Subject: IoT Hobby: Playing Online Games Age: 20 Birthdate: Nov 16, 2005	Ivan Villareal BSIT 3 ID: 2024-002 Email: ivan.villareal@edu.ph Favorite Subject: IoT Hobby: VideoGrapher Age: 19 Birthdate: Jan 31, 2002	 Tonyo Moreno BSIT 3 ID: 2024-004 Email: tonyo.moreno@edu.ph Favorite Subject: Project Management Hobby: Roblox Age: 18 Birthdate: Aug 4, 2005	Vincent Rebese BSIT 3 ID: 2024-006 Email: vincent.rebese@edu.ph Favorite Subject: Project Management Hobby: Playing Online Games Age: 21 Birthdate: Jan 5, 2005
 Ivan Villareal BSIT 3	 Joshua Santos BSIT 3 ID: 2024-005 Email: joshua.santos@edu.ph Favorite Subject: Project Management Hobby: CODE Age: 20 Birthdate: Sep 3, 2005	 Vincent Rebese BSIT 3 ID: 2024-006	 Wilken Nobetnov BSIT 3 ID: 2024-003 Email: wilken.nobetnov@edu.ph Favorite Subject: IoT Hobby: Gooning Age: 21 Birthdate: Mar 22, 2005

🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Student List CTF.

Use an appropriate commit message following proper Git standards.

Branch:

`feature/student-list`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
C:\Users\Manes Christopher\Documents\mannes\IT314AB_Artajo>git commit -m "complete student list CTF 4"
[feature/student-list 70125c2] complete student list CTF 4
4 files changed, 195 insertions(+), 152 deletions(-)
create mode 100644 my_first_flutter_app/assets/Binsoy.jpg
```

```
* feature/student-list
```

```
C:\Users\Manes Christopher\Documents\mannes\IT314AB_Artajo>git push origin feature/student-list
git: 'credential-manager-core' is not a git command. See 'git --help'.
Enumerating objects: 18, done.
Counting objects: 100% (18/18), done.
Delta compression using up to 4 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (10/10), 141.53 KiB | 9.43 MiB/s, done.
Total 10 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 5 local objects.
remote:
remote: Create a pull request for 'feature/student-list' on GitHub by visiting:
remote:      https://github.com/BosconianManes/IT314AB_Artajo/pull/new/feature/student-list
remote:
To https://github.com/BosconianManes/IT314AB_Artajo.git
* [new branch]      feature/student-list -> feature/student-list
```